

Cubic Surfaces, Statistically

Simon L. Rydin Myerson

2025-11-12

Cubic Surfaces

- ▶ **Cubic equations in three variables**

- ▶ For example

$$x^3 + y^3 + z^3 - 3 = 0.$$

Known integer solutions (Booker and Sutherland):

- ▶ (1, 1, 1)
- ▶ (4, 4, -5)
- ▶ (569 936 821 221 962 380 7203, -569 936 821 113 563 493 509, -472 715 493 453 327 032)
- ▶ **Very rare!**

Parametric Solutions

- ▶ Consider:

$$x^3 + y^3 + z^3 - 1 = 0.$$

- ▶ **Parametric solutions** exist, e.g.:

$$(9t^4, 3t - 9t^4, 1 - 9t^3).$$

- ▶ **Lots of solutions!**

General Goal

Find cubic surfaces with many integer solutions *not* coming from parametric solutions.

Specific Goals

- ▶ Survey a **lattice** generated by r cubic polynomials

$$g_1(x, y, z), \dots, g_r(x, y, z).$$

- ▶ Consider all linear combinations

$$f = a_1 g_1 + \dots + a_r g_r, \quad |a_i| < 5.$$

- ▶ Write code to find solutions to $f = 0$ with $|x|, |y|, |z| < 50$ and remove parametric solutions.
- ▶ Will be given written description of the algorithm.

Statistical Questions

- ▶ What is the distribution of the number of non-parametric solutions, when we choose a random f from the lattice?
- ▶ Is it different for different lattices?